

WINONA, MN, USA

WMO: 726588

Lat: 44.077N

Lon: 91.708W

Elev: 656

StdP: 14.35

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 04956

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-8.7	-4.1	-18.0	2.1	-4.9	-13.7	2.7	-1.8	22.0	18.6	19.3	19.6	3.8	300	0.482

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	90.7	73.4	88.4	72.9	84.1	70.6	77.1	87.2	74.9	84.0	73.0	81.4	7.1	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.5	127.6	82.0	72.2	122.3	80.5	70.3	114.3	78.5	40.8	87.5	38.8	84.2	37.0	81.7	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.6	16.4	14.0	DB	-17.0	96.6	7.9	6.4	-22.6	101.2	-27.2	104.9	-31.6	108.5	-37.3	113.2	(4)
			WB	-17.1	80.4	7.5	2.7	-22.5	82.3	-26.9	83.8	-31.1	85.3	-36.6	87.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	47.7	18.5	22.7	34.7	47.5	58.9	68.9	73.2	70.8	63.6	50.5	36.8	24.5	(6)
(7)		DBStd	21.18	12.38	12.09	11.42	9.70	8.25	6.69	5.85	5.78	8.10	9.16	9.60	11.03	(7)
(8)		HDD50	3716	977	765	491	158	16	0	0	5	106	407	791		(8)
(9)		HDD65	7052	1442	1185	940	529	224	34	5	12	121	458	846	1256	(9)
(10)		CDD50	2871	0	0	16	83	293	568	718	644	413	123	12	1	(10)
(11)		CDD65	731	0	0	1	4	36	152	258	191	79	10	0	0	(11)
(12)		CDH74	6411	0	0	4	68	375	1351	2311	1546	658	97	1	0	(12)
(13)		CDH80	1969	0	0	0	15	103	416	808	436	175	16	0	0	(13)
(14)	Wind	WSAvg	5.4	5.8	6.0	6.1	6.8	5.7	4.8	4.2	3.9	4.5	5.5	6.1	5.6	(14)
(15)	Precipitation	PrecAvg	34.6	1.1	0.9	1.9	3.8	4.0	4.7	4.3	4.9	3.7	2.1	2.0	1.3	(15)
(16)		PrecMax	42.8	2.8	2.6	3.3	7.8	9.9	9.3	8.9	12.9	10.0	6.6	6.7	2.5	(16)
(17)		PrecMin	24.5	0.4	0.2	0.4	1.1	1.6	1.9	1.3	1.0	0.9	0.7	0.2	0.3	(17)
(18)		PrecStd	5.3	0.7	0.6	0.9	2.1	1.7	2.2	1.9	2.6	2.2	1.4	1.5	0.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.3	54.8	72.2	81.4	90.0	93.1	96.9	93.5	90.4	81.7	70.0	54.4	(19)
(20)		MCWB	41.6	46.8	56.6	61.0	66.9	74.8	75.8	76.8	72.4	66.4	56.9	49.5	(20)	
(21)		2%	DB	41.0	48.2	63.4	73.4	81.9	89.8	91.0	89.7	84.4	76.6	61.3	46.2	(21)
(22)		MCWB	35.9	42.3	52.9	55.9	63.6	72.4	74.3	74.7	70.4	63.7	51.8	41.7	(22)	
(23)		5%	DB	36.7	42.8	56.9	69.9	78.6	84.4	89.5	84.1	81.1	71.7	55.2	41.3	(23)
(24)		MCWB	34.2	37.4	49.0	54.3	62.5	69.7	74.1	72.0	67.8	60.2	47.3	37.0	(24)	
(25)		10%	DB	34.4	37.4	51.9	63.4	73.2	81.3	83.8	81.5	77.3	65.7	51.9	37.1	(25)
(26)		MCWB	32.6	34.0	45.3	49.5	59.7	67.0	71.7	69.5	65.8	57.5	45.5	34.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	41.4	48.2	59.9	63.0	72.1	77.5	81.2	80.3	75.4	71.1	58.7	51.1	(27)
(28)		MCDWB	48.2	53.1	70.1	75.9	82.8	89.8	89.9	87.6	85.9	78.5	66.6	54.5	(28)	
(29)		2%	WB	36.9	42.5	54.9	58.7	68.0	74.7	77.9	76.3	72.6	65.1	53.3	42.1	(29)
(30)		MCDWB	40.2	48.6	61.1	70.9	77.7	85.2	88.1	84.8	81.4	71.5	59.7	45.8	(30)	
(31)		5%	WB	34.6	38.0	49.9	55.6	65.6	72.1	75.3	74.2	70.6	61.2	49.0	37.6	(31)
(32)		MCDWB	36.6	42.1	56.9	66.2	74.3	81.4	84.6	81.6	77.9	70.0	55.2	40.8	(32)	
(33)		10%	WB	32.7	34.8	44.5	52.2	62.8	69.9	73.2	72.2	68.3	57.5	45.6	34.9	(33)
(34)		MCDWB	34.3	37.8	51.0	62.2	71.1	77.9	81.8	79.0	75.2	64.7	50.6	37.1	(34)	
(35)	Mean Daily Temperature Range		MDBR	15.6	16.7	17.0	19.3	19.9	19.4	19.0	18.8	19.8	18.1	14.2	13.2	(35)
(36)		5% DB	MCDBR	15.9	19.3	24.2	28.6	25.9	23.4	22.0	21.8	23.5	25.1	22.0	16.9	(36)
(37)		MCWBR	13.1	14.2	14.7	14.6	11.9	10.4	9.7	9.9	11.3	14.0	14.4	13.3	(37)	
(38)		5% WB	MCDBR	14.7	17.9	22.7	24.1	21.0	19.8	19.4	18.6	19.8	21.4	20.4	15.8	(38)
(39)		MCWBR	12.6	13.6	15.0	14.3	11.7	10.4	10.1	10.1	11.0	14.0	14.8	13.0	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.265	0.297	0.325	0.370	0.403	0.410	0.411	0.406	0.375	0.330	0.295	0.268	(40)	
(41)		taud	2.395	2.306	2.373	2.315	2.270	2.289	2.292	2.317	2.381	2.496	2.506	2.442	(41)	
(42)		Ebn at Noon	277	283	290	283	276	273	271	269	269	270	264	265	(42)	
(43)		Edh at Noon	23	31	33	39	42	41	41	38	33	26	22	21	(43)	
(44)	All-Sky Solar Radiation	RadAvg	563	902	1226	1441	1696	1889	1966	1661	1324	849	553	408	(44)	
(45)		RadStd	56	90	134	146	131	146	84	110	95	114	49	39	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (49 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page